

ASTRODASHACARE

DIGITAL CONSULTING CENTER & ASTROLOGICAL COMPUTING ENGINE

Comprehensive End-to-End Application Workflow, Architectural Blueprint, Engine
Specifications & Technical Guide



USER JOURNEY

Kundli Generation, 20+ Astrological Engines, AI Guru, Live Consultations, Astro E-Commerce & Digital Wallet.



ASTROLOGER PORTAL

Call Queue Management, Live Native HUD & Kundli Viewer, Per-Minute Tariff, Break Timers & Wallet Earnings.



ADMIN CONTROL

Astrologer Approvals, Multi-CMS (Panchang, Ephemeris, Terms, Magazine), Mundane Global Analytics & RBAC.

Platform: Flutter Multi-Platform (Android / iOS / Web / Desktop)

Astrology Algorithms: Lahiri Ephemeris, KP System, BNN, Jamakol, Siddha Panchapakshi

Version: 2.0.0 Enterprise Release

Backend API: Python 3 FastAPI + MongoDB (with Persistent Storage Fallback)

Document Type: System Architecture & Operational Workflow Manual

Security & RBAC: JWT Token Auth, Guarded Role Routes, Dynamic Session Guard

1. EXECUTIVE SUMMARY & SYSTEM ARCHITECTURE

SYSTEM
DESIGN

AstroDashaCare (Astrocall) is an enterprise-grade digital astrological ecosystem combining ancient Vedic, Nadi, KP, and Tamil Siddha computational systems with modern real-time consulting, AI assistance, and commerce. The platform operates on a reactive multi-tier architecture enabling high-precision astronomical calculations alongside secure tele-consultations.

HIGH-LEVEL MULTI-TIER ARCHITECTURAL TOPOLOGY

LAYER	TECHNOLOGIES & COMPONENTS	CORE RESPONSIBILITIES
PRESENTATION LAYER	Flutter 3.x, Dart 3.11+, Google Fonts (Cinzel/Outfit), Flutter Animate, Shimmer, Svg	Reactive Glassmorphic UI, responsive layouts for Mobile/Web, real-time animation HUDs, South/North Indian chart renderers.
CLIENT COMPUTING	Dart Math Engines, Geocoding Database (10,000+ cities), Offline Ephemeris, PDF/Printing	Zero-latency calculation of 20+ astrological methods (SAV, KP Cusps, BNN, Horary, Panchapakshi), on-device PDF generation.
BACKEND SERVICE	Python 3 FastAPI, Pydantic, Uvicorn, CORS Middleware, RESTful API Suite	Centralized authentication, role verification, consultation signaling, CMS data ingestion, wallet billing, and database transactions.
PERSISTENCE LAYER	MongoDB Server (Production) + Mongomock with JSON File Sync (Offline Fallback)	ACID transactions for user profiles, astrologer verification records, magazine feeds, custom panchang, and wallet ledger.

THE THREE USER PERSONAS

1 Client / Seeker (Role: User)
Accesses daily horoscopes, personal kundli, 20+ calculation engines, AI assistant, joins call queues, buys spiritual products, and pays via wallet.

2 Certified Astrologer (Role: Astrologer)
Manages live availability, sets per-minute consultation tariffs, takes voice/video calls with live client kundli overlay, tracks payouts.

3 Platform Administrator (Role: Admin)

SECURITY & ROLE-BASED ACCESS CONTROL (RBAC)

The application enforces strict Route Guards (`_guardedRoute` in `main.dart`) and API authorization:

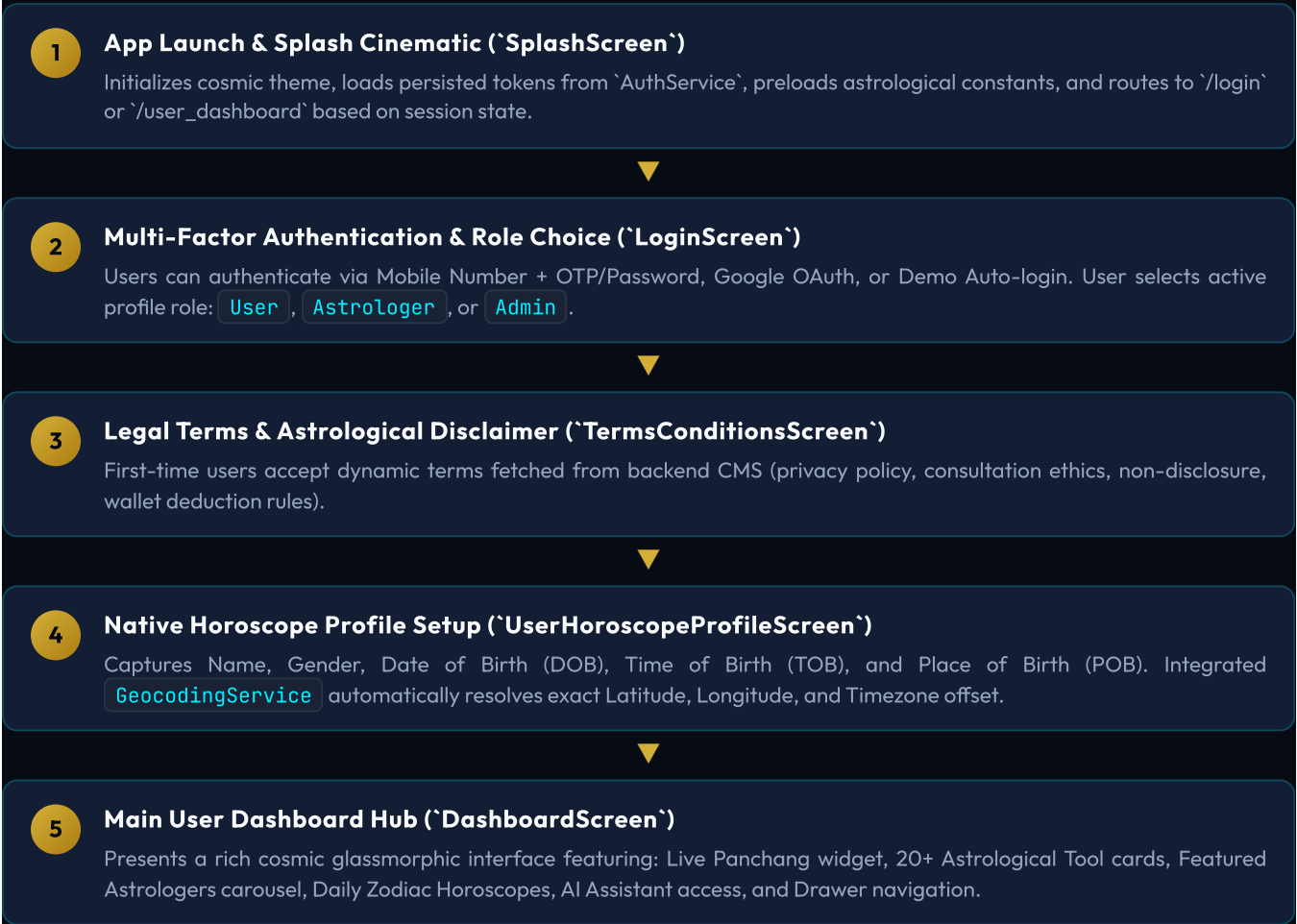
- Admin Lockdown:** Routes `/admin_dashboard`, `/admin_panchang`, `/admin_planet_positions`, `/mundane_astrology` reject non-admin sessions.
- Astrologer Gate:** `/astrologer_dashboard` requires verified Astrologer credentials.
- Anti-Contact Policy:** Phone numbers & bank data obfuscated during live sessions to protect platform integrity.
- Grace Refund:** Disconnections < 60s trigger automatic full wallet refunds.

Monitors system metrics, audits astrologer
KYC/credentials, manages Panchang &
Ephemeris CMS, edits T&C, and runs
Mundane analytics.

2. END-TO-END USER (CLIENT) WORKFLOW

USER EXPERIENCE

The Client Journey is designed to be intuitive, cosmic, and friction-free, guiding users from onboarding to in-depth astrological analysis and consultations.



☼ CLIENT FUNCTIONAL MATRIX & INTERACTION PATHS

 **KUNDLI & ANALYSIS**

CORE

Birth Chart & Calculations

Computes D1 Rasi, D9 Navamsha, Bhava Chalit, Vimshottari Dasha timeline, KP Cusps, BNN combinations, and Ashtakavarga matrices instantly.

 **LIVE CONSULTATION**

LIVE

Voice, Video & Chat

Selects certified astrologers, checks per-minute rate, joins call queue, engages in consultation with live wallet deduction, and rates astrologer post-call.

 **SHOP & WALLET**

COMMERCE

Remedies & Payments

Purchases certified natural gemstones, energized Rudrakshas, Yantras, recharges AstroDashaCare Wallet via Razorpay/UPI, and manages PDF settings.

3. ASTROLOGER PORTAL & ADMIN CONTROL WORKFLOWS

OPERATIONS

ASTROLOGER OPERATIONAL FLOW

1 Registration & Profile Verification
Submits bio, certifications, languages (Tamil, English, Hindi), experience years, specializations (Vedic, KP, Nadi, Jamakol), and desired per-minute rate.

2 Admin Approval Gate
Account placed in `Pending` status until Admin reviews credentials and activates the profile in the live roster.

3 Astrologer Dashboard (``AstrologerDashboardScreen``)
Controls live status: `OnLine`, `Busy`, or `Take Break` (with 5m/15m/30m countdown auto-revert).

4 Consultation HUD (``ConsultationScreen``)
Accepts incoming calls, displays client birth details, instant South Indian Rasi & Navamsha chart overlay, notes taker, and live earnings ticker.

5 Earnings & Payout Ledger
Real-time calculation of consultation revenues, platform revenue-share breakdown, and withdrawal requests.

ADMINISTRATOR CONTROL CENTER

1 Master Dashboard (``AdminDashboardScreen``)
Monitors high-level KPIs: Active Users, Verified Astrologers, Pending KYC Approvals, Total Consultations, and Platform Revenue.

2 Astrologer Roster Management
Add new astrologers, edit tariffs/specializations, upload avatars via `FilePicker`, approve/reject/delete astrologer profiles.

3 Daily Panchang CMS (``AdminPanchangCmsScreen``)
Overrides sunrise/sunset, Tithi, Nakshatra, Yoga, Karana, Rahu Kalam, Yamagandam, and festival dates for any given day.

4 Planetary CMS & Terms CMS
Manages live ephemeris coordinates, retrogrades, and legal disclaimers/terms through dynamic in-app CMS.

5 Mundane Astrology Monitor (``MundaneAstrologyScreen``)
Analyzes National Kundlis (e.g. India 15-Aug-1947), Mesha Sankranti Ingress, Eclipses, and Global Geopolitical trends.

 **Role Switching Feature:** The platform includes a seamless Workspace Switcher in `CosmicDrawer` allowing authorized administrators and astrologers to switch between Client View, Astrologer Workspace, and Admin Control Center without re-logging.

4. CORE ASTROLOGICAL ENGINES & CALCULATION WORKFLOWS

CALCULATION SUITE

AstroDashaCare houses the most comprehensive calculation suite in digital Vedic astrology, powered by optimized client-side Dart mathematical libraries and backend Swiss Ephemeris algorithms.

ASTROLOGICAL ENGINE	THEORETICAL BASIS & FORMULAS	KEY OUTPUTS & SCREEN DELIVERABLES
1. Birth Chart & Vargas (D1 to D60)	Lahiri Sidereal Ayanamsha, Sripathi / Placidus Bhava Cusps, Planetary Longitudes (Sun to Ketu).	South & North Indian Rasi & Navamsha charts, Padasaram table, Combustion & Retrograde indicators.
2. Vimshottari Dasha Engine	120-year Nakshatra Dasha cycle. Balance of birth dasha calculated from Moon's arc in Janma Nakshatra.	Mahadasha, Antardasha (Bhukti), Pratyantardasha, and Sookshma dasha start/end date timeline.
3. KP Astrology & Cusps	Krishnamurti Padhdhati (KP). 12 Placidus Cusps divided into Sign Lord, Star Lord, Sub Lord & Sub-Sub Lord.	12 Cusp Sub-lords, Planet Sub-lords, 4-Fold Significators table, and KP Ruling Planets HUD.
4. KP Horary (1 to 249)	Prashna seed number (1-249) maps directly to exact Zodiac arc & Ascendant Sub-lord.	Instant Horary Prashna Kundli, Query Significators, Sub-lord promise verification.
5. Bhrgu Nandi Nadi (BNN)	Sage Bhrgu & Nandikesha principles. Karaka planets (Guru=Jiva, Shani=Karma), 1-5-9 Trinal combinations, 2-12 links.	Directional Trine groupings (East, South, West, North), Life event predictions, Yoga analysis.
6. Jamakol Arudam	Ancient Tamil Prashna system. Day divided into 8 Jamams (3.75 Nazhigai each), Udhayam, Aarudam, Kavippu & 8 Jamakkol planets.	Complete Jamakkol Prashna Chakra, Kavippu danger point analysis, success/failure verdict.
7. Panchapakshi Shastra	Tamil Siddha 5-Bird Science (Vulture, Owl, Crow, Rooster, Peacock) by Sage Agathiyar. 5 states: Rule, Eat, Walk, Sleep, Die.	Day & Night Yamam activity charts, auspicious micro-periods (Arasu/Oon) for negotiations, deals, travel.
8. Marriage Porutham	10 & 12 Poruthams (Dina, Gana, Mahendra, Stree Dheerkha, Yoni, Rasi, Rasiyadhipathi, Vashya, Rajju, Vedha).	Total score out of 10/12, Sevvai (Manglik) Dosham check with exemptions, Rahu-Ketu Dosham, Papasamya score.
9. Ashtakavarga Matrix	8-fold benefic point contribution by 7 planets + Lagna. Sarvashtakavarga (SAV) & Bhinnashtakavarga (BAV).	SAV 337 points distribution chart, Trikona Shodhana, Ekadhipatya Shodhana, Pinda Shodhana.
10. Live Hora & Dina Suddhi	Planetary horas based on local Sunrise/Sunset. 24 unequal horas per solar day with cyclic planetary rulers.	Real-time active Hora countdown, favorable/unfavorable activity guide, Dina Suddhi 21-Dosha audit.
11. Pindayu Longevity	Classical Ayurdaya calculations. Identifies Maraka (2/7), Badhaka houses, Harana (reductions) for combust/enemy planets.	Alpayu (0-32), Madhyayu (33-66), Deerghayu (67-100+) lifespan classification & longevity score.

ASTROLOGICAL ENGINE	THEORETICAL BASIS & FORMULAS	KEY OUTPUTS & SCREEN DELIVERABLES
12. Numerology & Lo Shu	Chaldean & Pythagorean systems. Root Number (Moolank), Destiny (Bhagyank), Name Number, 3x3 Lo Shu Grid.	Lo Shu Element plane analysis (Mind, Soul, Practical), Lucky numbers, colors, gems, compatible dates.
13. Tamil Calendars	Tamil Solar Year (60-year cycle), Tamil Months (Chithirai to Panguni), Paksha, Nalla Neram, Rahu/Gulika Kalam.	Daily Panchang card, Monthly Tamil Calendar view, Gowri Panchangam, Subha Muhurtham dates.
14. Tara Balam & Chandrashtama	9 Tara classifications (Janma, Sampat, Vipat, Kshema, Pratyak, Sadhana, Naidhana, Mitra, Param Mitra) across 27 stars.	Daily Nakshatra strength index, Chandrashtama warning indicator (Moon in 8th from Janma Rasi).

5. LIVE CONSULTATION, CALLING & BILLING ENGINE

COMMUNICATION

AstroDashaCare provides a seamless, secure tele-consultation pipeline linking Clients with certified Astrologers with dynamic per-minute billing and automatic wallet deduction.

CONSULTATION SESSION LIFECYCLE FLOW

1

Astrologer Selection & Rate Inspection

User navigates to the Consult Tab, filters by specialization/language/rating, and inspects per-minute tariff (e.g., ₹30/min).



2

Pre-Call Wallet Sufficiency Validation (`PaymentService` & `WalletScreen`)

System verifies user has minimum required balance (at least 5 minutes worth of consultation). If insufficient, a fast top-up dialog is triggered via UPI / Razorpay.



3

Queue Entry & Signaling Handshake (`CallingService`)

Call request sent to backend FastAPI signaling hub. The Astrologer's device receives an interactive incoming call popup with client birth coordinates.



4

Active Consultation HUD (`ConsultationScreen`)

Upon connection, a real-time call timer initiates. Astrologer views a split-screen HUD showing client's live Rasi chart, current Mahadasha, notes panel, and mute/camera controls.



5

Dynamic Billing & Wallet Settlement

Wallet balance is deducted per minute. If balance runs out, audio prompt alerts the user with a 60-second grace window to recharge before auto-termination.



6

Post-Call Review & Rating Dialogue (`RatingDialog`)

Client submits 1-5 star rating and written testimonial. Astrologer's public rating and consultation count update automatically.

WALLET & PAYMENT ARCHITECTURE

- **Supported Payment Methods:** Razorpay Gateway, UPI (GPay, PhonePe, Paytm), NetBanking, Credit/Debit Cards.
- **Wallet Top-Up Packs:** ₹100, ₹250, ₹500, ₹1,000, ₹5,000 with promotional bonus cash.
- **60-Second Grace Rule:** Calls disconnected < 60 seconds due to network drops receive a 100% refund.
- **Ledger Transparency:** Downloadable invoice statement with exact timestamps and per-second duration.

ASTRO SHOP E-COMMERCE ENGINE

- **Certified Gemstones:** Natural Ruby, Yellow Sapphire, Blue Sapphire, Emerald, Diamond, Red Coral, Hessonite.
- **Energized Rudraksha:** 1 to 14 Mukhi laboratory-certified beads.
- **Sacred Yantras:** Copper, Silver & Gold-plated Sri Yantra, Kuber Yantra, Navagraha Yantra.
- **Checkout Flow:** In-app Cart → Address Entry → Wallet or Gateway Payment → Tracking ID.

6. PDF GENERATION ENGINE & REST API DIRECTORY

INTEGRATION

PROFESSIONAL PDF HOROSCOPE ENGINE (` PdfGeneratorService`)

AstroDashaCare contains an on-device PDF generation subsystem producing publication-grade, multi-page Tamil and English Sidereal Horoscope reports:

CUSTOM BRANDING

Configurable Astrologer Center Name, Subtitle, Contact numbers, Address, and Custom Invocation Slokam in [CustomerDetailsSettingsScreen](#) .

VECTOR CHARTS

Crisp vector rendering of South Indian Rasi & Navamsha grids, planetary degrees, retrograde markers, and Pada distributions.

UNICODE TAMIL FONTS

Integrated [PdfFontManager](#) supporting Noto Sans Tamil & Anek Tamil for rendering complex Tamil ligatures.

FASTAPI RESTFUL API ENDPOINTS DIRECTORY

METHOD	ENDPOINT URI	ACCESS LEVEL	DESCRIPTION
POST	/api/auth/register	Public	Register new user or astrologer
POST	/api/auth/login	Public	Authenticate via phone/password
POST	/api/auth/google	Public	Google OAuth 2.0 verification
GET	/api/astrologers	Public	Fetch verified astrologer list
POST	/api/astrologers	Admin	Create new astrologer entry
DELETE	/api/astrologers/{id}	Admin	Remove astrologer from system
GET	/api/panchang/today	Public	Get daily panchang calculations
PUT	/api/panchang/entry	Admin	Admin Panchang CMS update
GET	/api/terms	Public	Fetch active Terms & Conditions
PUT	/api/terms	Admin	Update dynamic platform policies
POST	/api/wallet/topup	User / Admin	Credit funds to user wallet
POST	/api/wallet/deduct	System	Per-minute consultation billing
GET	/api/magazine/feed	Public	Fetch Astrogen Magazine articles
POST	/api/magazine/article	Admin	Publish new magazine article

🌟 **Offline Resilience:** The backend API includes an automatic in-memory `mongomock` engine with file-backed JSON serialization (`astrocall_mongodb_data.json`), ensuring that the system functions seamlessly even when external MongoDB servers are offline.

7. SUMMARY, TECHNICAL HIGHLIGHTS & ROADMAP

ROADMAP

✦ KEY ARCHITECTURAL HIGHLIGHTS

🚀 PERFORMANCE & SCALABILITY

All 20+ astrological calculation routines run locally on client devices with near-zero latency, avoiding expensive round-trips for core mathematical charts.

🎨 COSMIC GLASSMORPHIC AESTHETICS

Custom dark theme with tailored HSL tokens (Deep Space `#050914`, Radiant Gold `#FFD700`, Cyber Cyan `#00F0FF`, Cosmic Violet `#8A2BE2`).

🔒 FAULT-TOLERANT PERSISTENCE

Dual-strategy persistence ensuring zero data loss during network outages with persistent JSON disk state serialization.

🤖 AI-AUGMENTED ASTROLOGY

Deep integration with AI Assistant models contextualized with native Kundli placements for instant remedial insights.

COMPLETE ROUTE NAVIGATION INDEX

NAMED ROUTE	DART SCREEN CLASS	GUARDED ROLE	MODULE PURPOSE
/	SplashScreen	Public	Cinematic Splash & Bootstrapping
/login	LoginScreen	Public	Multi-Role Authentication & OAuth
/terms	TermsConditionsScreen	Public	Mandatory Terms Acceptance
/profile	UserHoroscopeProfileScreen	Public	Birth Chart Profile Builder
/user_dashboard	DashboardScreen	User / Any	Client Home Hub & Services Grid
/astrologer_dashboard	AstroLogerDashboardScreen	Astrologer	Live Call Queue & Workspace
/admin_dashboard	AdminDashboardScreen	Admin	Master CMS & Roster Management
/admin_panchang	AdminPanchangCmsScreen	Admin	Daily Panchang CMS Editor
/admin_planet_positions	AdminPlanetPositionsCmsScreen	Admin	Ephemeris & Coordinates CMS
/admin_terms	AdminTermsCmsScreen	Admin	Legal Policy & Disclaimer CMS
/birth_chart	BirthChartScreen	Public	D1 to D60 Kundli & Padasaram
/kp_astrology	KpAstrologyScreen	Public	KP 12 Cusps & Sub-Lord System
/kp_horary	KpHoraryScreen	Public	1-249 Horary Prashna Engine
/bhrigu_nandi_nadi	BhriguNandiNadiScreen	Public	BNN Trine & Directional Combinations
/jamakol_arudam	JamakolArudamScreen	Public	Tamil Jamakkol Prashna Chakra
/panchapakshi	PanchapakshiScreen	Public	Siddha 5-Bird Auspicious Timings
/marriage_porutham	MarriagePoruthamScreen	Public	10/12 Poruthams & Dosha Matching
/ashtakavarga	AshtakavargaScreen	Public	SAV & BAV 337-Point Matrices

NAMED ROUTE	DART SCREEN CLASS	GUARDED ROLE	MODULE PURPOSE
/hora	LiveHoraScreen	Public	Live Planetary Hora Schedule
/nazhigai	NazhigaiScreen	Public	Udayadhi Nazhigai Timekeeping
/longevity	LongevityScreen	Public	Classical Pindayu Longevity Estimator
/daily_calendar	DailyCalendarScreen	Public	Daily Panchang, Tithi & Gowri Time
/tamil_month_calendar	TamilMonthCalendarScreen	Public	Tamil Monthly Calendar & Festivals
/numerology	NumerologyScreen	Public	Chaldean & Pythagorean Numerology
/tara_balam	TaraBalamScreen	Public	9-Tara Strength & Chandrashtama
/ai_assistant	AiHoroscopeScreen	Public	AI Astrological Chatbot Guru
/shop	ShopScreen	Public	Gemstones, Rudrakshas & Yantras
/wallet	WalletScreen	User / Admin	Wallet Recharge, History & Ledger
/magazine	MagazineFeedScreen	Public	Astrogen Digital Magazine & Articles
/pdf_settings	CustomerDetailsSettingsScreen	Public	Custom PDF Header/Footer Settings
/mundane_astrology	MundaneAstrologyScreen	Admin	National Kundli & Geopolitics